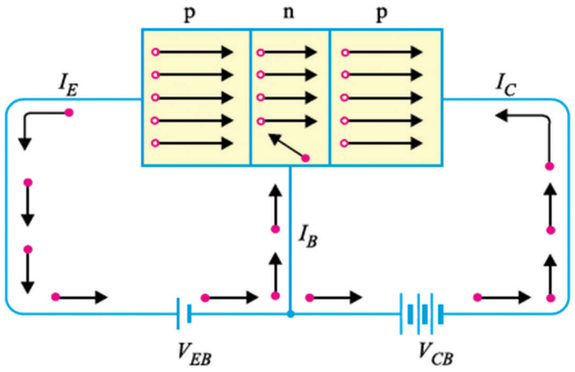


Fig 10.10 *p-n-p Transistor*

holes 5% will combine with the electrons. The remainder 95% cross into the collector region to constitute collector current I_C. In this way, almost the entire emitter current flows in the collector circuits. It may be noted that current conduction within pnp-transistor is by holes. However, in the external connecting wires, the current is still by electrons.

10.6 LIGHT EMITTING DIODE (LED)

The Light emitting diode(LED) is a PN junction device which emits light when forward biased. In all semiconductor PN junctions, some of the energy will be radiated as heat and some in the form of photons. In silicon and germanium, greater percentage of energy is given out in the form of heat and the emitted light is insignificant. In other materials such as gallium phosphide (Gap) or gallium arsenide phosphide (Ga As P) the number of photons of light energy emitted is sufficient to create a visible light source. Here, the charge carrier recombination takes place when electrons form the n-side cross the junction and recombine with the holes on the p-side.

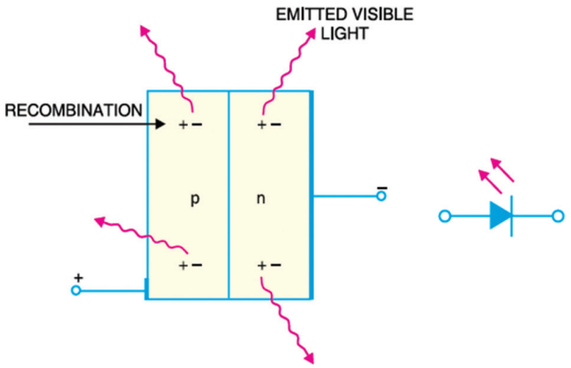


Fig 10.11 *LED Display*

LED under forward bias and its symbol are shown in the Fig 10.11. When an LED is forward biased, the electrons and holes move towards the junction and recombination takes place. As a result of recombination, the electrons lying in the conduction bands of n-region fall into the holes lying in the valance band of a p-region. The difference of energy between the conduction band and the valance band is radiated in the form of light energy. Each recombination causes radiation of light energy. Light is generated by recombination of electrons and holes where by their excess energy is transferred to an emitted photon. The brightness of the emitted light is directly proportional to the forward bias event.

Applications

- LEDs can be switched 'on' and 'off' at a very fast speed of one nano second (1ns).
- They are used in burglar alarm systems, picture, phones, multimeters, calculators, digital meters, microprocessors, digital computers, etc.

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